



## Interval Estimation of Mean ( $\sigma$ Known & Unknown) Transcript

Hello everyone, in this part of our module on estimation and hypothesis testing, we are focussing on a core concept, interval estimation of the mean. Over the next 10 or so minutes, we are going to go beyond making a single guess about a population's average. We will learn more robust methods of constructing a range of credible values.

We will explore how to do this in two distinct scenarios, one in which we actually know population variability and the one in which we do not know population variability or what we know as sigma. These are two things that we are going to look into. So let's get started.

Let's begin by discussing why a single number or a point estimate often isn't enough. When we want to understand a large population, say the average height of all the students in a university, we can't measure everyone, right? So what we do, we take a sample, the average height of our sample, which we call  $\bar{x}$  is a point estimate. It is our single best guess for the true average.

But here's a crucial issue. If we take a different sample, we would get a different sample mean. This natural chance driven variation from sample to sample mean, our point estimate will always never be perfectly correct, right? The difference between our sample mean and the true population mean  $\sigma \mu$  is called sampling error.

A point estimate on its own is a shot in the dark. It does not tell us how much sampling error we might have. This is where interval estimate or confidence interval becomes essential.

It provides a range of values, what we believe likely contains a true population mean. Think of it this way, a financial analyst, say here, a financial analyst is predicting earnings of a share, right? And the prediction could very well could be a simple point estimate, which is in this case 1.5 share. And that might not be the very right value, right? If the analyst gives a forecast, which is in range, let's say the price per share is going to be in between 1.4 to 1.6, that could be a more credible, more reliable estimate because there are more chances that the value itself would lie in between this range, right? So building a confidence interval is itself a real valued because there are more chances that true estimate of the population would belong in that range.

And that's the entire essence of doing all of these things and moving beyond single guess. Okay. So how do we construct this range? Thankfully, the structure is very logical and consistent.

Every confidence interval is built around point estimate here and we have sample mean. So we have a point estimate, we do plus minus of the margin of error. And hence we get interval estimate.



Now we are going to learn about what exactly is this margin of error and how do we constitute, what constitutes this margin of error? Okay. So you can actually think of margin of error as a radius of uncertainty. It depends on how confident we are and how the entire variability in our data is.

Our methods for calculating it hinges around critical question. Is the population standard deviation  $\sigma$  is known or unknown? So if we know population standard deviation, this margin of error might be very limited and it might not be a good or it might not be into ranges. But when you know  $\sigma$  or you do not know  $\sigma$ , you are actually doing an or finding an estimate from the sample, which is where things become a little critical.

But we will let you know about different, different types of methods, what you do in those situations. Okay. So case one, when  $\sigma$  is known, let's explore the first scenario, an ideal but the real situation where population standard deviation  $\sigma$  is known.

This could happen in a controlled manufacturing setting, for example, where a machine has been producing a part for years and its output variability is extremely well documented. Okay. When  $\sigma$  is known, the central limit theorem tells us that the distribution of sample means will be normal.

This allows us to use the standard normal distribution or Z distribution to build our estimate. And that's why you see here, you have sample mean, the estimate, and you have plus minus the entire margin of error. Now this margin of error, if you see, is constituted of two parts, one, which is  $Z_{\alpha/2}$ , the standard Z distribution, and then the standard error,  $\sigma/\sqrt{n}$ . So the entire margin of error, which I, which we discussed in the last slide, is constituted of two things, one, the critical value, which is derived from the Z distribution, and then the sampling standard error.

Okay. So, that's what it is. Now, the very first thing, the critical value  $Z_{\alpha/2}$  is a number which comes directly from the Z distribution, and it is determined by the chosen level of confidence.

If we want a 95% confidence interval, we need to find Z scores that capture the central 95% of the distribution. This leaves two points. So if you think of, let's say, here, if you think of 95% distribution, let me hide the t distribution.

So 95% could be very well in between, where am I highlighting, from this region to this region. Okay. And if that constitutes 95%, it also means that there would be some area which would be here on these two sides, right? So a higher confidence interval, let's say 99, would require a larger Z value.



Because in that case, 99 could, the left-hand side, lower bound of 99% would be here, while the upper bound might be here, right? I just wanted to give you how it looks like on a graph. But that's what it is. You have critical value  $Z_{\alpha/2}$ , which is depicted from Z distribution, and then we have a standard error of the mean, which is  $\sigma/\sqrt{n}$ . So these two things constitute our margin of error.

And once we know both of them, we can very well constitute the entire confidence interval around the mean. Okay. Now there are situations wherein  $\sigma$  is unknown, and these situations are more practical in nature, right? You will encounter almost every single time in practise that  $\sigma$  is unknown.

If we do not know the population mean, it's highly improbable that we would know it's a standard deviation, and that's what the main issue is, right? So we have to estimate the entire standard deviation. And how do we do that? So what we do is, we use sample estimate, or just sample standard deviation to estimate population standard deviation. So you see here, you have  $\sigma$ , which is population standard deviation, which is unknown.

So what we do is, we estimate it using a sample standard deviation, which is  $S$ . The problem now entirely is, when you take different different samples, your sample standard deviation would also vary, right? And that's when this distribution, t-distribution, arrives. So there was an individual named as William C. Gossett, who published his wonderful work under pseudonymy student, and he developed the student's t-distribution. The formula for t-interval looks remarkably similar to what we have for the z-interval.

We simply swap the values of our estimate  $S$  with  $\sigma$ . If you see here, the only difference that we have in this formula, and what we had previously, if you see, right, we have, in place of  $z$ , we have  $t$ , and then we have  $\alpha/2$ , which is the level of significance, then we have degrees of freedom, okay? Because I told you we are estimating the population standard deviation using sample standard deviation, and in t-test, it takes into account the concept of degrees of freedom, which is nothing but total number of samples that we have, minus 1, okay? Now and basically this is the sample size minus 1, that's how we get 2 degrees of freedom. This is also introduced in Bessel's correction, the formula that we use for variance, if you all know.

The formula for variance is how much deviation do we have from the mean, divided by  $n-1$ . So sample standard deviation is a formula which takes into account  $n-1$ , and that  $n-1$  is nothing but coming from the concept of degrees of freedom, okay? So the main thing, let's get back to the main thing, the main thing is, when  $\sigma$  is unknown, in that case, we estimate the entire confidence level, sorry, the entire range, okay, confidence interval, using the t-test and sample standard deviation. This formula is very much similar to what we have for the margin of error when  $\sigma$  is known, in



that case, instead of  $T_{\alpha/2}$  degrees of freedom, we use  $Z$ , and we use, instead of  $S$ , we use  $\sigma$ , because  $\sigma$  is known in that case.

So the idea here is, when the population standard deviation is unknown, use a sample standard deviation to, for, in that place, right? So now let's visualise why we need this different distribution. The blue line here, if you see, is the familiar  $Z$  distribution, and the orange dashed line represents a  $T$  distribution, this one, okay? Notice that the  $T$  distribution is a shorter and has fatter tail. If you see here, it is shorter and also, it might become more fat, more variability, okay? It would have, when I say a fatter tail, it means that it might have more variability.

They, both of these, they represent a higher probability of observing values far from the mean. This is the uncertainty penalty we must pay for having to estimate  $\sigma$  by using  $S$ . If you see here, in this case, because in  $T$  distribution, we are using sample standard deviation, right, to estimate population standard deviation, hence this entire range, that's why it becomes more broader, okay? By having fatter tails, the  $T$  distribution forces us to use a larger critical value, okay, which results in a wider, more appropriated cross-years confidence interval. However, let's say, for example, if our sample size  $N$  gets larger, our estimate, this entire distribution, the  $T$  distribution, if you see, if our sample size grows and if we take more and more sample size, in that case, the entire  $T$  distribution, approximately, you will see that it would become very similar to what you have as a blue curve or the standard normal distribution.

So, they are very similar, it's just that when the sample size grows, in that case, it would become similar to what we have as normal distribution. So, you can see here, this is the entire normal distribution. When we plot  $T$  distribution, it might not look, it is still very similar because there is, the only variability which is coming in is because we are estimating sample, sorry, population standard deviation using the samples, right? When we have larger and larger sample, you can also think of it as law of large numbers, in that case, our variability would be lower, right? And hence, this entire  $T$  distribution, it becomes very much like what we have here, the blue distribution which is the normal distribution, standard normal distribution, okay? Now, this is an interactive test that I wanted all of us to look into.

Now, let's put all of these things that we have learned into practise with this simulation. The vertical red line, if you see here, right, this line represents the true population mean and that's what has been written out here, if you see, okay? This is the value that we are trying to capture. In real life, this is always hidden from us, right, because we generally do not know what exactly is true population mean.

Each horizontal line, when I will draw the samples, you will see each horizontal line here somewhere, they would represent the entire range or what we say as confidence interval. These are calculated from random samples. So, let's start the exploration with



the default setting, let's say, 95% confidence interval and a sample size of, let's say, 30.

Let's go ahead and draw 100 samples. Now, what do you see here? You see 100 different intervals here or let's say, you have point estimate which are in black colour or dark colour and then you have an entire interval estimated around it. So, you have a point estimate and an entire interval.

We have drawn 100 samples, that's why you see 100 lines. Now, what do you see uniquely or what is unique to this entire simulation? Majority of your interval actually contains the true mean, this red line that you see, right. There are cases, let's say, this particular case wherein I am hovering, there are cases when the entire interval estimation does not contain the true mean but they are very less, right.

In this case, there are actually 5. We have 95% confidence interval and what does it mean? It means that when we take 100 samples, there are exactly 95 cases in this particular example. There are exactly 95 cases when our true population estimate is actually lying in the ranges that we are constituting, right, in all of those samples. Let's say, we take another sample.

You will see that because we are taking 95% confidence interval, majority of the ranges that will be constituting around the point estimates in all of those samples, they would majorly capture true population mean and that's why the entire range, that's why the example that I gave you about the shares initially, right. You understand that when you say the forecast for any share is going to be in this range, there are more chances that the true value belongs to that range and that's what this is depicting. Now, what would happen if you change sample sizes? Let's say, I have a very small sample size.

Let's say, I am taking a lot of samples but the sample size is low. In that case, if I draw, now you see that when I have a smaller sample size, my entire range increases, right. Now, the entire interval is huge, right and when I take a large sample size, the range becomes narrower.

So, we are extremely extremely confident when we take a sample with more sample size or we take samples with more sample size. You see here, the entire range becomes more narrower. We are more confident and there are also chances that we will make lesser mistakes.

Herein, we have only two mistakes if you see, right. So, this is what would happen. Now, what if we have a certain sample size but we change confidence interval? You see our range or the entire estimate, the confidence interval that we build or the entire range of values that we build, they become unrealistic or they become unreliable, okay and that's what would happen in majority of the cases.



So, this is what I wanted to help you understand. I hope all of you have got the understanding of what happens here. So, let's summarise the entire thing that we have gone through, okay.

The purpose is to build an interval estimate for the mean that provides a range of credible values. I have told you repeatedly in this entire lecture that this is what we do generally because our entire range of values might have greater probability of containing the true population estimate that we are trying to uncover, right. So, the critical question is if our standard deviation on the population is known or unknown.

If the standard deviation is known, we use the margin of error as  $Z$  into the standard error  $\sigma$  by  $\sqrt{n}$ . If we do not know, then in that case, we use  $t$ -distribution, right. So, if we know sample standard deviation, it's well and good. In that case, we will use  $Z$ -distribution.

If we do not know, then we will use  $t$ -distribution. That's how simple it is. Then we looked into that simulation if you just remember, right.

Increasing the confidence level, let's say from 95 to 99 makes the interval wider and that's what in practise you will also assume, right. In real scenarios, that's what would happen. And increasing the sample size makes the interval more narrower.

That's what you also saw in that simulation if you remember, right. So, this is what I wanted to cover. Hope this was all okay and you are able to relate to the concept.

-----End-----